HKUST – Department of Computer Science and Engineering
COMP 2711: Discrete Math Tools for Computer Science

Fall 2023 Midterm Examination

Date: 31 October 2023 Time: 19:00–21:00

Name: _____	Student ID: _____
Email: _____	Lecture: _____

Instructions

- This is a closed book exam. It consists of **18** pages and **9** questions. You are allowed to bring an A4 one-page single-sided handwritten cheat sheet.
- Please write your name, student ID, email, lecture section in the space provided at the top of this page.
- Please sign the honor code statement on page 2.
- Answer all the questions within the space provided on the examination paper. You may also use the back of the pages. Each question is on a separate page. This is for clarity and is not meant to imply that each question requires a full-page answer. Some can be answered using only a few lines.
- Solutions can be written in terms of binomial coefficients, factorials, and the $C(n,k)$, $P(n,k)$ notation. For example, you can write $\binom{5}{3}+\binom{4}{2}$ instead of 16. Avoid using nonstandard notations such as ${}_nP_k$ or ${}_nC_k$. Calculators are not allowed for the exam.

Problems	1	2	3	4	5	6	7	8	9	Total	B
Points	11	12	7	16	7	12	11	8	16	100	3
Score											

As part of HKUST's introduction of an honor code, the HKUST Senate has recommended that all students be asked to sign a brief declaration printed on examination answer books that their answers are their own work, and that they are aware of the regulations relating to academic integrity. Following this, please read and sign the declaration below.

I declare that the answers submitted for this examination are my own work.

I understand that sanctions will be imposed, if I am found to have violated the University regulations governing academic integrity.

Student's Name: _____

Student's Signature: _____

Problem 1: Propositional Logic [11 pts]

(a) [4 pts] Prove the following propositional equivalence **without** using a truth table:

$$(p \to r) \vee (q \to r) \equiv (p \wedge q) \to r$$

(b) [4 pts] Let p, q, r, s, t denote five propositions. You are given the following set of premises:

1. $p \to (q \to r)$
2. $p \vee s$
3. $t \to q$
4. $\neg s$

Using rules of inference for propositional logic, show that the truth of the above premises implies that the conclusion $\neg r \to \neg t$ is true.

(c) [3 pts] Determine if the two compound propositions $(p \to q) \to (r \to s)$ and $(p \to r) \to (q \to s)$ are logically equivalent. Show a proof if they are, or give a counter-example if they are not.

Answer: (a) $(p \to r) \vee (q \to r) \equiv (\neg p \vee r) \vee (\neg q \vee r) \equiv \neg p \vee \neg q \vee r \equiv (\neg p \vee \neg q) \vee r \equiv \neg(p \wedge q) \vee r \equiv (p \wedge q) \to r$

(b)

1.	$\neg s$	Premise
2.	$p \vee s$	Premise
3.	p	(1,2)
4.	$p \to (q \to r)$	Premise
5.	$q \to r$	(3,4)
6.	$t \to q$	Premise
7.	$t \to r$	(5,6)
8.	$\neg r \to \neg t$	(7)

(c) Not equivalent. Counter-example: $p = \mathrm{F}, q = \mathrm{T}, r = \mathrm{F}, s = \mathrm{F}$

Marking scheme: (a) 4 points for correct proving steps tracing from (LHS to RHS) or (RHS to LHS). -1 points for each incorrect intermediate step of proof.

(b) 4 points for completing all proving steps from one proposition to another. -1 point for each incorrect use of inference rule.

(c) 3 points for correct answer of 'Not Equivalent' with correct proof or correct truth table.
+2 points for correct answer of 'Not Equivalent' with unreasonable argument on the proof.

+1 point for correct answer of 'Not Equivalent' with incorrect proof or incorrect truth table.
+1 point for correct answer of 'Not Equivalent' without proof and without truth table.

Problem 2: Predicate Logic [12 pts]

(a) [4 pts] Express the negation of the following statement so that all negation operators immediately precede the predicates:

$$\exists x(P(x) \to \forall y(Q(y) \to (R(y) \vee S(y))))$$

(b) [4 pts] Let the domain consist of all integers. $CoPrime(a, b)$ is a predicate which means a and b are co-prime (you do not need to know the meaning of 'co-prime' in order to answer this question). Express the following statement using quantifiers:
Two numbers a and b are co-prime if and only if any integer can be expressed as a linear combination of a and b.
(A linear combination is an expression constructed from a set of terms by multiplying each term with a constant and summing the results (e.g., a linear combination of x and y is any expression of the form $ax + by$, where a and b are constants).)
Note: you are only allowed to use quantifiers, addition, multiplication, equality, and the *CoPrime* predicate.

(c) [4 pts] Let the domain consist of all people in a pub and $P(x)$ denote the statement that x is drinking. Briefly argue if this statement is always true or not.

$$\exists x\,(P(x) \to \forall y\,P(y))$$

(There is someone in the pub such that, if he or she is drinking, then everyone in the pub is drinking.)

Answer: (a)

$$\begin{aligned}
& \neg\exists x(P(x) \to \forall y(Q(y) \to (R(y) \vee S(y)))) \\
\equiv\ & \forall x\neg(\neg P(x) \vee \forall y(\neg Q(y) \vee (R(y) \vee S(y)))) \\
\equiv\ & \forall x(P(x) \wedge \neg\forall y(\neg Q(y) \vee R(y) \vee S(y))) \\
\equiv\ & \forall x(P(x) \wedge \exists y\neg(\neg Q(y) \vee R(y) \vee S(y))) \\
\equiv\ & \forall x(P(x) \wedge \exists y(Q(y) \wedge \neg R(y) \wedge \neg S(y))) \\
\equiv\ & \forall x P(x) \wedge \exists y(Q(y) \wedge \neg R(y) \wedge \neg S(y))
\end{aligned}$$

(b) $\forall a\forall b\,(CoPrime(a, b) \iff \forall c\,\exists x\exists y\,(c = ax + by))$

(c) This is always true. For a better understanding, we convert the implication to disjunction: $\exists x\,(\neg P(x) \vee \forall y\,P(y))$. If everyone in the pub is drinking, then this statement is true because $\forall y\,P(y)$ is true. If not everyone is drinking, then there exists someone who is not drinking and the statement is again true.

Marking scheme: (a) 4 points for correct conclusion.
+1 point for incorrect conclusion but can show at least some logical and reasonable steps.
-1 point for each incorrect proving step.
-1 point for incorrect quantifier.
-1 point for negation operator not broken down to each predicate.

(b) 4 points points for correct translation of the predicate (the formula).
-1 point for each incorrect quantifier.
-1 point for each incorrect logical operator.
-1 point for missing assignment of c variable where Forall(c) and There-Existed(x,y) (c = ax + by).

(c) 4 points for correct answer of "Always True" with sufficient argument.
+3 points for correct answer of "Always True" with insufficient argument.
+2 points for correct answer of "Always True" with incorrect argument or missing argument.

Problem 3: Proof by Contradiction, Sets, Functions, Cardinality [7+3 pts]

(a) [4 pts] Let S and T denote two sets. Prove the following by contradiction:
If $S \cap T = \emptyset$ and $S \cup T = T$ then $S = \emptyset$.

(b) [3 pts] Without the need for providing any explanation, determine if the following sets are infinitely countable (C) or infinitely uncountable (U):

(i) [1 pts] The union of countably many countable sets

(ii) [1 pts] The set of all functions from $\{1, 2\} \to \mathbb{N}$

(iii) [1 pts] The set of all functions from $\mathbb{N} \to \{1, 2\}$

(c) [**Bonus** 3 pts] An algebraic number is a number that is a root of a non-zero polynomial in one variable with integer coefficients. As an example, $(1+\sqrt{5})/2$ is an algebraic number, because it is a root of the polynomial $x^2 - x - 1$. Or, the complex number $i + 1$ is an algebraic number as it is a root of $x^4 + 4$. Briefly argue why we have countably many algebraic numbers.

Hint: Try to think about how many polynomials exist and how many roots each one has. You may use the result of any of (b) as well.

Answer: (a) Assume that S is not an empty set. This implies that $\exists x \; s.t. \; x \in S$. If $x \in S$, then $x \in S \cup T$ or $x \in T$ (because $S \cup T = T$). However, $x \in S \cap T$ contradicts with the fact that $S \cap T = \emptyset$. Consequently, we must have $S = \emptyset$.

(b) C, C, U

(c) You can express the set of algebraic numbers as a countable union of finite sets.

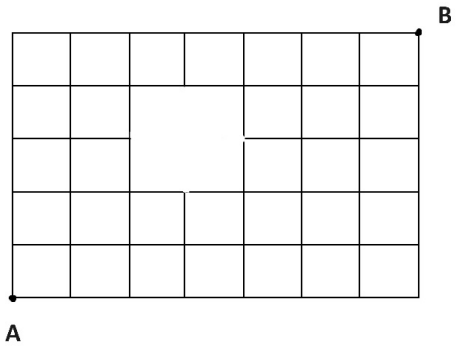
Marking scheme: (a) 0.5 point for setting up the contradiction
1 point if you can use the intersection to find that element is also in T.
2.5 points for finding the contradiction.

(b) 3 points

(Bonus) 1 point if you can construct the polynomial form.
1 point for saying that the set of polynomials with integer coeficient is countable.
1 more point for conclusion.

Problem 4: Counting and Pigeonhole Principle [16 pts]

(a) Consider the following "incomplete" 5×7 grid. We are standing at position A and we want to go to position B. We are only allowed to go right or up. In how many ways can we go from A to B?



(b) Consider the following 3×3 grid. You are sitting in your car on the bottom left cell and the aim is to reach the top right cell (shown by a star) as shown in Figure 1a. You want to color **every** cell in this 3×3 grid **except** the top right cell with colors either Blue(B) or Red(R) (i.e., you will color 8 cells). If your car is on a Red cell you move one cell to the right and if your car is standing on a Blue cell you move one cell up. How many different colorings exist such that you can reach to the top right cell? Note that you need to write down the combinatorial format and explain your solution.

As an example, Figure 1b shows a valid coloring as you will reach to the top right cell as shown. However, Figure 1c shows an invalid coloring as you will not be able to reach to the top right cell.

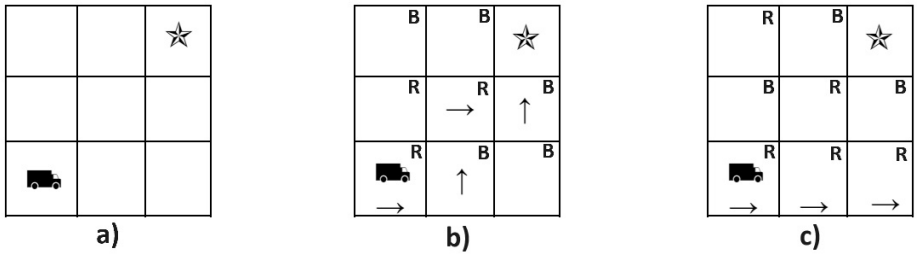
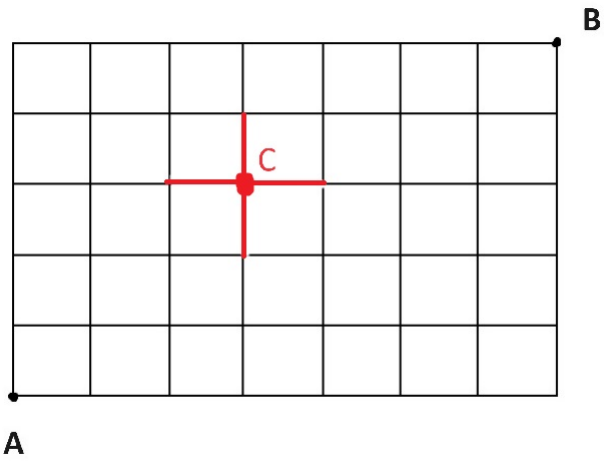


Figure 1: R/B denote the red/blue colors of the cells

(c) Imagine having a 4×4 grid. We want to write either $+1$ or -1 in its cells such that the sum of every row is equal to 0. Similarly we want the sum of every column to be 0 also. How many distinct "valid" tables exist?

(d) Imagine having an equilateral triangle with each side being 3 cm. Prove that for all possible ways that you can place 10 points inside this equilateral triangle, there exist at least two points with their distance being less than 1 cm.

Answer: (a) Basically the question is the same as asking how many ways to go from A to B without passing through C, which can be calculated as $\binom{12}{5} - \binom{6}{3} \cdot \binom{6}{2}$.



(b) Notice that if you consider any path, the coloring of that path is unique. We have $\binom{4}{2}$ different paths from bottom left to top right with a unique coloring. There will be 4 cells left that can accept any color and it does not effect. So the final answer is $\binom{4}{2} \cdot 2^4$.

(c) The first row needs two $+1$ and two -1. So there are $\binom{4}{2}$ different ways to complete the first row. For the second row there are 3 general cases:

Case 1: The columns of the second row have the exact same values as row one. For example, if row one is $\{+1, +1, -1, -1\}$ then the second row is also $\{+1, +1, -1, -1\}$. This case happens only in one valid permutation of the second row. In this case row 3 and row 4 have only a unique filling.

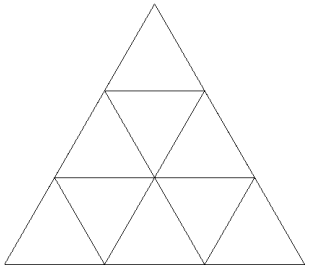
Case 2: All columns of the second row have different values with row one. For example, if row one is $\{+1, +1, -1, -1\}$ then the second row is $\{-1, -1, +1, +1\}$. This case happens only in one valid permutation of the second row. In this case row 3 can accept any valid input which has $\binom{4}{2}$ different ways and the last row has a unique filling.

Case 3: Half of the columns of the second row have different values with row one and half of them are the same. For example, the first

row is $\{+1, +1, -1, -1\}$ and the second row is $\{-1, +1, +1, -1\}$. This case happens in four different permutations of the second row. In this case there are 2 ways to complete row 3 and row 4 (why?).

So the total number of solutions will be $\binom{4}{2} \cdot (1 \cdot 1 + 1 \cdot 6 + 4 \cdot 2)$.

(d) We split the equilateral triangle of sides 3 into 9 equilateral triangles of side 1cm as follows:



Now we want to put 10 points **inside** this triangle. Which means at least two points will be either inside one of the $1 \times 1 \times 1$ triangle cells (based on pigeonhole principle and that there are 10 points and 9 triangles) which means their distance is less than 1. Also it is possible for both points to be on the internal lines (not on the border lines though) but even in that case the results remains the same because the distance between any two points on the internal lines (and not on the border lines) are still less than 1 and also if one point is one the internal line and another point is completely inside the $1 \times 1 \times 1$ related triangle, their distance is still less than 1cm.

Marking scheme: (a) 2 points for getting the total number of ways from A to B by assuming C exists.
2 points for getting the total number of ways from A to B by not passing C.

(b) 2 points for getting the number of different paths with a unique color.
2 points for getting the number of cells that can accept any color that does not affect.

(c) 1 point for dividing the problem into 3 cases. For each correct answer in each case, the student gets 1 point.

(d) 2 points for splitting the triangle into 9 equilateral triangle of sides 1.
0.5 points to show how to do it.
1.5 points for the pigeonhole principle argument.
(Some small partial credits might have been given to other solutions)

Problem 5: Combinatorial Proof [7 pts]

Prove the following identity using combinatorial arguments:

$$\sum_{i=0}^{n} \binom{n}{i}^2 = \binom{2n}{n}$$

Answer: It is equivalent to proving the identity $\sum_{i=0}^{n} \binom{n}{i} \cdot \binom{n}{n-i} = \binom{2n}{n}$.

Let us consider a set of $2n$ elements, and we want to select n elements from this set. The right-hand side (RHS) of the equation directly gives us the total number of ways to do this, which is $\binom{2n}{n}$.

Now, let us focus on the left-hand side (LHS). We divide the set into two parts, each containing n items. We then select items for each part. If we select i items for the first part, we must select $n - i$ items for the second part. Therefore, there are $\binom{n}{i} \cdot \binom{n}{n-i}$ ways to make these selections.

To consider all possible values of i, we sum up these quantities for i ranging from 0 to n. This gives us the sum $\sum_{i=0}^{n} \binom{n}{i} \cdot \binom{n}{n-i}$.

Hence, both the RHS and LHS represent the number of ways to select n elements from a set of $2n$ elements, and thus, the equation holds true.

Marking scheme: 2 points for illustrating correctly with the RHS
1 points for illustrating correctly with the LHS
2 points for the key process $\sum_{i=0}^{n} \binom{n}{i}^2 = \sum_{i=0}^{n} \binom{n}{i} \cdot \binom{n}{n-i}$
2 points to prove LHS and RHS are the same argument

Problem 6: Inclusion and Exclusion [12 pts]

How many solutions are there for the equation $a+b+c = 13$, where a, b, c are nonnegative integers and $a \leq 4, b \leq 5, c \leq 6$. Use the inclusion-exclusion principle to calculate the result instead of enumerating the values of possible solutions.

Answer:

$$\begin{aligned} U &= \{(a,b,c) \,|\, a+b+c = 13,\ a,b,c \geq 0\} \\ A &= \{(a,b,c) \,|\, a+b+c = 13,\ a > 4, b, c \geq 0\} \\ B &= \{(a,b,c) \,|\, a+b+c = 13,\ b > 5, a, c \geq 0\} \\ C &= \{(a,b,c) \,|\, a+b+c = 13,\ c > 6, a, b \geq 0\} \end{aligned}$$

So the result of our question is $|\overline{A} \cap \overline{B} \cap \overline{C}| = |U| - |A \cup B \cup C|$. And we use inclusion-exclusion to count $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$.

We know that the number of r-combinations from a set with n elements when repetition of elements is allowed is $\binom{n+r-1}{r}$, which means that the number of solutions of $a + b + c = n$, where $a, b, c \geq 0$, is $\binom{n+2}{2}$.

$$|U| = \binom{15}{2} = 105$$

Since we can rewrite the set A to be

$$A = \{(a,b,c) \,|\, a+b+c = 13,\ a > 4, b, c \geq 0\} = \{(a'+5,b,c) \,|\, a'+b+c = 8,\ a', b, c \geq 0\}$$

$$|A| = \binom{10}{2} = 45$$

In the same way, we can calculate the sizes of B, C as well as their intersections.

$$\begin{aligned} |B| &= \binom{9}{2} = 36 \\ |C| &= \binom{8}{2} = 28 \\ |A \cap B| &= \binom{4}{2} = 6 \\ |A \cap C| &= \binom{3}{2} = 3 \\ |B \cap C| &= \binom{2}{2} = 1 \end{aligned}$$

While the set $A \cap B \cap C$ should satisfy $a+b+c = 13$ where $a > 4, b > 5, c > 6$, there is no solution for this equation. So we have,

$$|A \cap B \cap C| = 0$$

In all, the result is,

$$\begin{aligned} & |\overline{A} \cap \overline{B} \cap \overline{C}| \\ = \; & |U| - (|A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|) \\ = \; & 105 - (45 + 36 + 28 - 6 - 3 - 1 + 0) \\ = \; & 6 \end{aligned}$$

Marking scheme: 3 points for setting up the definition of U, A, B, C.
4 points for getting the key expressions right for $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$ (only get the points if you set up the definition of A, B and C correct.)
2 points for getting the right answer of $|U|$, $|A|$, $|B|$, $|C|$ in combination expression.
2 points for getting the right answer of $|A \cap B|, |A \cap C|, |B \cap C|, |A \cap B \cap C|$.
1 point for getting the final answer right.
Note:
(1) You can do it directly or by complement as shown in the answer. Both answers will be accepted if it is right.
(2) We will not accept an answer that hasn't been well explained or is not in combination form. For example, get the answer by counting the possible cases. Only 1 point will be given even if the answer is right.

Problem 7: Counting and Probability [11 pts]

Imagine the communication infrastructure of HKUST contains n antennas in a line. Due to a typhoon, m of them are broken at random and are not functioning. The whole communication system remains functional if no two consecutive antennas are broken. What is the probability that the communication system remains functional?

Answer: The total number of ways that m antennas are broken out of n antennas is equal to $\binom{n}{m}$. Now let us enumerate the malfunctioning antennas $1, 2, \ldots, m$. We define x_i as the number of unharmed antennas between i and $i+1$, such that x_0 corresponds to the number of unharmed antennas before the first broken antenna and x_m corresponds to the number of unharmed antennas after the last broken antenna. We know that $x_0 + x_1 + x_2 + \cdots + x_m = n - m$. The constraints are $x_0, x_m \geq 0$ and $x_i > 0 \quad (0 < i < m)$. After changing variables according to their constraints we get $y_0 + y_1 + y_2 + \cdots + y_m = n - 2m + 1$ for $(y_i \geq 0)$. This equation has $\binom{n-m+1}{m}$ solutions. So the probability that the communication system remains functional is $\frac{\binom{n-m+1}{m}}{\binom{n}{m}}$.

Marking scheme: 2 points for the total number of ways.
4 points for the final value of number of cases that the communication system remains functional.
5 points for explanation on how to calculate it.
(Some partial credits might have been given to other type of solutions)

Problem 8: Conditional Probability [8 pts]

Two computers, a sender and a receiver, are communicating with each other. The sender transmits bit 1 with a probability of 30% and bit 0 with a probability of 70%. The communication channel is noisy, and we know the following information about the transition of data:

1. If the sent bit is 0, there is a 90% probability that the receiver will receive bit 0 and a 10% probability that it will receive bit 1.
2. If the sent bit is 1, there is an 80% probability that the receiver will receive bit 1 and a 20% probability that it will receive bit 0.

Given that the receiver computer received bit 0, determine the probability that the bit being sent was also 0.

Answer: Denote the bit sent by the sender to be S and the bit received by the receiver to be R.

We know that:
$p(S = 1) = 0.3,\ p(S = 0) = 0.7$
$p(R = 1 \mid S = 1) = 0.8,\ p(R = 0 \mid S = 1) = 0.2$
$p(R = 1 \mid S = 0) = 0.1,\ p(R = 0 \mid S = 0) = 0.9$

By the Bayes' theorem,

$$
\begin{aligned}
& p(S = 0 \mid R = 0) \\
= & \frac{p(R = 0 \mid S = 0)\, p(S = 0)}{p(R = 0)} \\
= & \frac{p(R = 0 \mid S = 0)\, p(S = 0)}{p(R = 0 \mid S = 1)\, p(S = 1) + p(R = 0 \mid S = 0)\, p(S = 0)} \\
= & \frac{0.9 \times 0.7}{0.2 \times 0.3 + 0.9 \times 0.7} \\
= & \frac{21}{23} \\
= & 0.913
\end{aligned}
$$

Marking scheme: 8 points

1) You will get 2 points if you successfully conclude the know conditional probabilities.

2) You will then get 2 points if you then show the correct expression of Baye's theorem.

so, 4 points if you know Baye's theorem.

3) You will then get 3 points if you write the correct expression: $\frac{0.9 \times 0.7}{0.2 \times 0.3 + 0.9 \times 0.7}$

4) The final calculation accounts for 1 point, if you get 21/23 or 63/69 or 0.63/0.69, you can earn this 1 point.

Incorrect conditional probabilities and calculation may cause lose of points.

Problem 9: Random Variables [16 pts]

Each time, a robot can move X cm to the right (negative X means the robot move to the left). X is a discrete random variable taking values from $\mathbb{N}$ with the following PMF (probability mass function),

$$p(X=i) = \begin{cases} \frac{2}{3} & \text{if } i=1 \\ \frac{1}{3} & \text{if } i=-1 \\ 0 & \text{otherwise} \end{cases}$$

(a) [2 pts] Calculate $E(X)$ and $V(X)$.

(b) [2 pts] If $Y = X - 1$, calculate $E(Y)$ and $V(Y)$.

(c) [2 pts] If $Z = X(X-1)$, calculate $E(Z)$.

(d) [10 pts] This robot is standing at position 0. It stops once it reaches to position +1 cm. Before stopping, the robot moves T times where T is a random variable.

(i) [1 pts] When k is a positive integer, calculate $P(T = 2k)$.

(ii) [7 pts] When k is a positive integer, calculate $P(T = 2k-1)$.

(iii) [2 pts] What is the expected value of T?

Answer: (a)

$$\begin{aligned} E(X) &= 1 \cdot \frac{2}{3} - 1 \cdot \frac{1}{3} \\ &= \frac{1}{3} \\ E(X^2) &= 1 \cdot \frac{2}{3} + 1 \cdot \frac{1}{3} \\ &= 1 \\ V(X) &= E(X^2) - E(X)^2 = \frac{8}{9} \end{aligned}$$

(b)

$$\begin{aligned} E(Y) &= E(X-1) = E(X) - 1 = -\frac{2}{3} \\ V(Y) &= V(X-1) = V(X) = \frac{8}{9} \end{aligned}$$

(c)

$$E(Z) = E(X(X-1)) = E(X^2 - X) = E(X^2) - E(X) = \frac{2}{3}$$

(d) Denote every right move to be R and every left move to be L. Since the robot will stop once it is in position +1 cm, the last move can only be from 0 to +1 cm. Thus, the last move is R, and before the last move, we need to have the same number of R and L. Denote the subsequence excluding the last move to be sequence S. L is legal if when we read from left to right, we have always encountered at least as many L as R.

(i) Since S has the same number of R and L, the length of S must be even. Adding the last move R, it is impossible for T to be an even number. Thus $P(T = 2k) = 0$

(ii) $P(T = 2k-1) = \frac{2}{3}^k \cdot \frac{1}{3}^{k-1} \cdot (\binom{2k-2}{k-1} - \binom{2k-2}{k})$
We know from the above that we need to distribute $k-1$ L's and k R's in the sequence, and the last item must be R. There are $\binom{2k-2}{k-1}$ ways to form S which containing $k-1$ R's and $k-1$ L's. However, we need to exclude the "illegal" cases. Here we prove that the number of illegal cases is equal to the number of possible sequences of $(k-1)+1$ L's and $(k-1)-1$ R's, which is $\binom{2k-2}{k}$
Suppose S is illegal. Then there is a least i where there is a R at position i and equally many L's and R's before i. Now swap all L's to R's and all R's to L's in the first i positions of S. This gives us a possible sequence of $(k-1)+1$ L's and $(k-1)-1$ R's. Conversely, suppose we are given a sequence M of $(k-1)+1$ L's and $(k-1)-1$ R's, and let i be the first position where there are more L's up to that point than R's. Flipping those R's and L's gives us back a sequence of k L's and k R's, which is illegal, because there are more R's up to i than L's.

(iii) $E(T) = \sum_{i=1}^{\infty} i \cdot P(T = i) = \sum_{k=1}^{\infty}(2k-1) \cdot P(T = 2k-1) = \sum_{k=1}^{\infty}(2k-1) \cdot \frac{2}{3}^k \cdot \frac{1}{3}^{k-1} \cdot (\binom{2k-2}{k-1} - \binom{2k-2}{k}) = 3$

Marking scheme: (a) 2 points

(b) 2 points

(c) 2 points

(d) 10 points

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